

Connect 4 Use Case Descriptions

Use Case 1: Drop a Piece

Iteration: 1

Primary Actor: Player

Goal in context: The player drops a piece into a column on the board

Preconditions:

1. Column must be empty (must have at least 1 available slot)
2. It's the player's turn
3. The game is in progress

Trigger: The player selects a column to drop a piece

Scenario:

1. The player selects a column
2. The system checks if the column is empty
3. If so, the system places the piece in the lowest available row of the column
4. Then it switches to the other player's turn

Postconditions:

1. The board updates the new piece placement
2. Switches to other players turn

Exceptions: The system notifies the player if the column is full and stops the move

Priority: High - It's a primary action in the game

When available: From the start of the game

Frequency of use: Multiple times per game - every turn

Channel to actor: Game interface

Secondary actors: None

Channel to secondary actors: N/A

Open issues: None

Use Case 2: Check for Win

Iteration: 1

Primary Actor: System

Goal in context: The system verifies if a player has won the game

Preconditions: At least 1 move has been made

Trigger: A piece is placed on the board

Scenario:

1. The system checks if four pieces of the same colour exist in a row, column, or diagonal, in a row
2. If four pieces in a row are met, the system declares the current player as the winner
3. The game ends

Postconditions: The winner is announced and the game ends

Exceptions: If the system fails to detect the win, the game continues wrongly (bug)

Priority: High - Checks when the game ends

When available: After any move is made

Frequency of use: Multiple times per game - every turn

Channel to actor: Game interface

Secondary actors: None

Channel to secondary actors: N/A

Open issues: None

Use Case 3: Check for Draw

Iteration: 1

Primary Actor: System

Goal in context: The system checks if the game ends in a draw

Preconditions:

1. All board slots are filled
2. No player has won - the winning condition has not been met by either player

Trigger: The board becomes filled after the last move

Scenario:

1. The system checks if all columns are full
2. The system confirms no player has achieved four in a row
3. If both checks are met, the system declares the game a draw
4. The game ends

Postconditions: The game ends with no winner

Exceptions: If the system fails to detect a draw, the game may continue wrongly (bug)

Priority: High - Needed for handling end-game conditions

When available: At the end of a game if no winner is found

Frequency of use: Once per game (If all columns are full with no win condition)

Channel to actor: Game interface

Secondary actors: None

Channel to secondary actors: N/A

Open issues: None

Use Case 4: Check for Game Over

Iteration: 1

Primary Actor: System

Goal in context: The system determines if the game should end after a move is made

Preconditions: A player drops a piece on the board

Trigger: Triggered automatically whenever a move is completed

Scenario:

1. The system calls "Check for Win" to determine if the current player has won
2. If a win is detected, the system declares the current player the winner and ends the game
3. If no win is detected, the system calls "Check for Draw" to see if the board is full
4. If the board is full and no player has won, the system declares a draw
5. If both a win and a draw are not detected, the game continues

Postconditions:

1. If a player wins, the game ends with a winner
2. If no player has won and the board is full, the game ends in a draw
3. If neither condition is met, the game continues

Exceptions: If the system fails to check for a win or draw correctly, the game may continue wrongly (bug)

Priority: High - Determines when the game stops, making essential

When available: After each move

Frequency of use: Once per turn

Channel to actor: Game system processes

Secondary actors: None

Channel to secondary actors: N/A

Open issues: None

Use Case 5: Forfeit

Iteration: 1

Primary Actor: Player

Goal in context: The player forfeits the game, letting the other player win

Preconditions: The game must be in progress

Trigger: The player selects "Forfeit Game"

Scenario:

1. The player selects "Forfeit Game"
2. The system checks to confirm the action
3. The opponent wins

Postconditions: The game ends, and the opponent wins

Exceptions: If the system fails to detect the forfeit, the game may continue wrongly (bug)

Priority: Medium - Helpful for player experience but not needed for normal gameplay

When available: Anytime during a game

Frequency of use: Low

Channel to actor: Game menu interface

Secondary actors: None

Channel to secondary actors: N/A

Open issues: None